ASSIGNMENT

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4MW23CS040
A Section

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1)Write a Java program to demonstrate String creation, comparison, substring(), length(), toUpperCase(), and StringBuffer append().
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Summary:

Java program demonstrates String operations like creation, comparison, substring(), length(), and toUpperCase().
It also uses StringBuffer to append text, showing mutable string behavior.

Code:

```
import java.util.Scanner;

public class Main {

    @SuppressWarnings("resource")
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```java
System.out.print("Enter a string (only alphabets allowed): ");
String input = sc.nextLine();

// Validation: only alphabets
if (!input.matches("[a-zA-Z]+")) {
    System.out.println("Invalid input! Only alphabets are allowed.");
    return;
}

// String Creation
String str1 = input;
String str2 = new String(input);

// Comparison
System.out.println("\n--- String Comparison ---");
System.out.println("Using == : " + (str1 == str2));
System.out.println("Using equals(): " + str1.equals(str2));

// Length
System.out.println("\nLength of string: " + str1.length());

// Substring
if (str1.length() >= 3) {
    System.out.println("Substring (1,3): " + str1.substring(1, 3));
} else {
    System.out.println("Substring not possible (string too short)");
}

// toUpperCase
System.out.println("\nUppercase: " + str1.toUpperCase());

// StringBuffer append
StringBuffer sb = new StringBuffer(str1);
sb.append("_JAVA");
System.out.println("StringBuffer result: " + sb);
```

```
    }
}
```

Output:

```
Enter a string (only alphabets allowed): Volleyball

--- String Comparison ---
Using == : false
Using equals(): true

Length of string: 10
Substring (1,3): ol

Uppercase: VOLLEYBALL
StringBuffer result: Volleyball_JAVA
```

```
Enter a string (only alphabets allowed): 12
Invalid input! Only alphabets are allowed.
```

2)Create a Student class and store objects using ArrayList. Perform insertion, traversal, and sorting using Comparator.
->

Summary:
Student objects are stored in an ArrayList with insertion and traversal using loops.Sorting is performed using Comparator based on attributes like name or marks.

Code:
```
package com.assignment;

import java.util.*;

// Student class
class Student {
    int id;
    String name;
    int marks;

    Student(int id, String name, int marks) {
        this.id = id;
        this.name = name;
        this.marks = marks;
    }
}

// Comparator for sorting by name
class NameComparator implements Comparator<Student> {
    public int compare(Student s1, Student s2) {
        return s1.name.compareTo(s2.name);
    }
}
```

```java
public class Assignment2Collections {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        ArrayList<Student> list = new ArrayList<>();

        System.out.print("Enter number of students: ");
        int n = sc.nextInt();

        // Insertion (user input)
        for (int i = 0; i < n; i++) {
            System.out.println("\nEnter details of student " + (i + 1));

            System.out.print("ID: ");
            int id = sc.nextInt();

            System.out.print("Name: ");
            String name = sc.next();

            System.out.print("Marks: ");
            int marks = sc.nextInt();

            list.add(new Student(id, name, marks));
        }

        // Traversal
        System.out.println("\nBefore Sorting:");
        for (Student s : list) {
            System.out.println(s.id + " " + s.name + " " + s.marks);
        }

        // Sorting
        Collections.sort(list, new NameComparator());

        // After sorting
```

```java
        System.out.println("\nAfter Sorting (by Name):");
        for (Student s : list) {
            System.out.println(s.id + " " + s.name + " " + s.marks);
        }

        sc.close();
    }
}
```

Output:

```
Enter number of students: 3

Enter details of student 1
ID: 1
Name: Akshay
Marks: 95

Enter details of student 2
ID: 2
Name: Dinesh
Marks: 86

Enter details of student 3
ID: 3
Name: Harish
Marks: 73

Before Sorting:
1 Akshay 95
2 Dinesh 86
3 Harish 73

After Sorting (by Name):
1 Akshay 95
2 Dinesh 86
3 Harish 73

Process finished with exit code 0
```

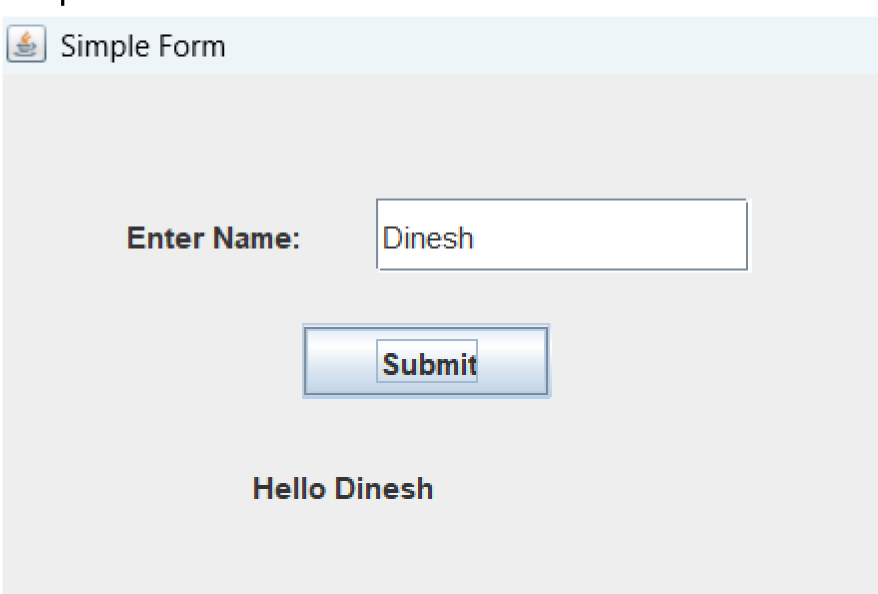
3)Design a GUI form using JLabel, JTextField, and JButton.
Display the entered name when the button is clicked.
->

Summary:
A GUI form is designed using JLabel, JTextField, and JButton components.
On button click, the entered name is displayed using ActionListener.

Code:
```
package com.assignment;

import javax.swing.*;
import java.awt.event.*;

// Swing GUI class
public class Assignment3Swing {
    public static void main(String[] args) {

        // Create frame
        JFrame frame = new JFrame("Simple Form");

        // Components
        JLabel label = new JLabel("Enter Name:");
        JTextField textField = new JTextField();
        JButton button = new JButton("Submit");
        JLabel result = new JLabel("");

        // Set positions (x, y, width, height)
        label.setBounds(50, 50, 100, 30);
        textField.setBounds(150, 50, 150, 30);
        button.setBounds(120, 100, 100, 30);
        result.setBounds(100, 150, 200, 30);

        // Button click event
        button.addActionListener(new ActionListener() {
```

```
            public void actionPerformed(ActionEvent e) {
                String name = textField.getText();
                result.setText("Hello " + name);
            }
        });

        // Add components to frame
        frame.add(label);
        frame.add(textField);
        frame.add(button);
        frame.add(result);

        // Frame settings
        frame.setSize(400, 300);
        frame.setLayout(null);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

Output:



4)Develop a HelloServlet program that displays the message 'Welcome to Advanced Java Lab' using Tomcat deployment.
->

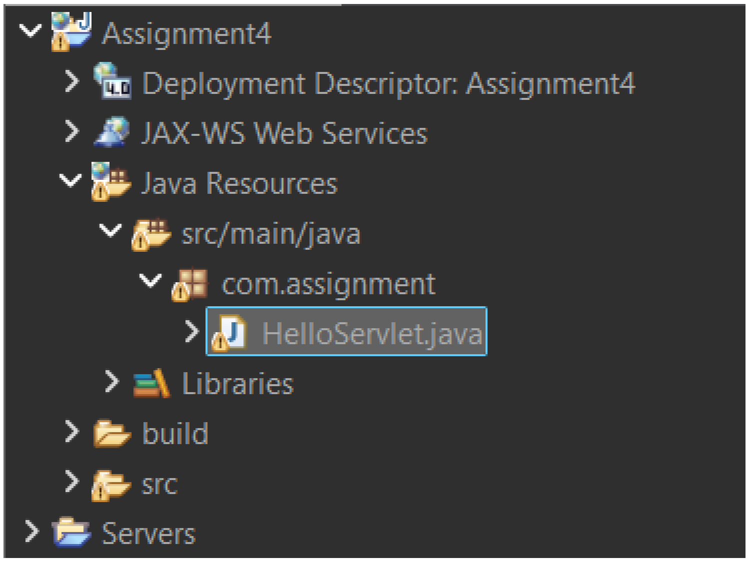
Summary:

HelloServlet is created using HttpServlet and deployed on Apache Tomcat server.It displays the message "Welcome to Advanced Java Lab" in the browser using doGet() method.

Code:
HelloServlet.java

```
package com.assignment;

import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/hello")
public class HelloServlet extends HttpServlet {

   protected void doGet(HttpServletRequest request, HttpServletResponse
response)
         throws IOException {

      response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      out.println("<h2>Welcome to Advanced Java Lab</h2>");
   }
}
```

Project Structure:



Output:

5)Write a JDBC program to connect Java to MySQL database and execute SELECT * FROM student to display USN, Name, and Branch.

->
Summary:

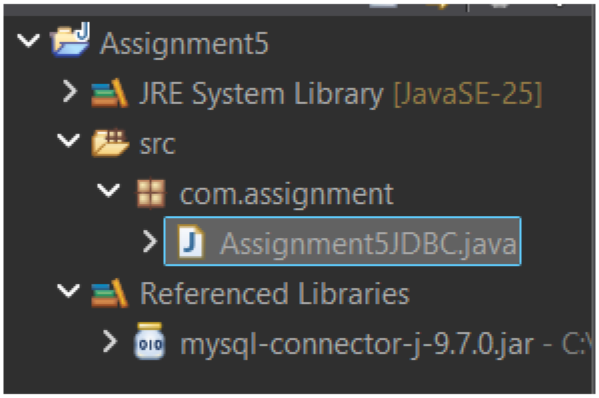
JDBC is used to connect Java application with MySQL database.It executes SELECT query to retrieve and display USN, Name, and Branch from student table.

Code:
Assignment5JDBC.java

```
package com.assignment;

import java.sql.*;

public class Assignment5JDBC {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/college";
        String user = "root";       // change if needed
        String password = "your_password";   // your MySQL password

        try {
            // Load MySQL driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Establish connection
            Connection con = DriverManager.getConnection(url, user,
password);
```

```
            // Create statement
            Statement stmt = con.createStatement();

            // Execute query
            ResultSet rs = stmt.executeQuery("SELECT * FROM student");

            // Display result
            System.out.println("USN\tName\tBranch");
            while (rs.next()) {
                String usn = rs.getString("usn");
                String name = rs.getString("name");
                String branch = rs.getString("branch");

                System.out.println(usn + "\t" + name + "\t" + branch);
            }

            // Close connection
            con.close();

        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Folder Structure:



Output:

```
USN       Name      Branch
1BM21CS001          Ganesh   CSE
1BM21CS002          Rahul    CSE
1BM21CS003          Ankit    ISE
```